

Stony Brook University
Alejandro Melo Ponce

Spring 2018

Midterm 2 Take-home exam

Due: Friday, May 4, before 11:59PM through BlackBoard.

INSTRUCTIONS: You are encouraged to work with your classmates but each student **must write his or her own solutions**. You must typeset your solutions in any word processor of your choice. Please also attach the R code that you use to solve the problems. I will open up a link in Blackboard in which you will be able to submit your files. **For all of the problems, be sure to use robust standard errors. I will grade this exam using three problems. If you choose to do a fourth one, it will count as extra. The choice of which three problems to solve is yours.**

Notation: Recall that $\hat{\beta}$ denotes the estimate of the unknown parameter β ; $\log(x)$ is the *natural logarithm* of the variable x ; Δx denotes the change in the variable x , i.e. $\Delta x = x' - x$; finally, $\% \Delta x$ denotes the percentage change in x , i.e. $\% \Delta x = 100 \frac{\Delta x}{x}$.

1. Use the data in WAGE1 for this problem.

- (a) Use OLS to estimate the equation

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{exper}^2 + u$$

and report the results using the usual format.

- (b) Is exper^2 statistically significant at the 1% level?
 - (c) Using the approximation

$$\% \Delta \widehat{\text{wage}} \approx 100(\hat{\beta}_2 + 2\hat{\beta}_3 \text{exper}) \Delta \text{exper},$$

find the approximate return to the fifth year of experience¹. What is the approximate return to the twentieth² year of experience? Put a 95% confidence interval around the predicted returns at both the fifth and twentieth year of experience.

- (d) At what value of exper does additional experience actually lower predicted $\log(\text{wage})$? How many people have more experience in this sample? *Hint: Check problem 4d in PS5.*

2. Use the data in VOTE1 for this exercise, which contains data on election outcomes and campaign expenditures for 173 two-party races for the U.S. House of Representatives in 1988. Let voteA be the percentage of the votes received by candidate A, expendA and expendB the campaign expenditures by candidate A and B respectively, and partystrA is a measure of party strength for Candidate A (the percentage of the most recent presidential vote that went to A's party).

- (a) Consider a model with an interaction between expenditures:

$$\text{voteA} = \beta_0 + \beta_1 \text{partystrA} + \beta_2 \text{expendA} + \beta_3 \text{expendB} + \beta_4 \text{expendA} \times \text{expendB} + u$$

What is the partial effect of expendB on voteA , holding partystrA and expendA fixed? What is the partial effect of expendA on voteA ? Is the expected sign for β_4 obvious?

¹That is, going from 4 years to 5 years.

²Again, this is going from 19 to 20 years.

- (b) Estimate the equation in part (a) and report the results in the usual form. Is the interaction term statistically significant?
- (c) Find the average of $expendA$ in the sample. Fix $expendA$ at 300 (for \$300,000). What is the estimated effect of another \$100,000 spent by Candidate B on $voteA$? Is this a large effect?
- (d) Now fix $expendB$ at \$100. What is the estimated effect of $\Delta expend = 100$ on $voteA$? Does this make sense? Estimate a 99% confidence interval around the predicted change on $voteA$.
- (e) Now, estimate a model that replaces the interaction with $shareA$, candidate A's percentage share of total campaign expenditures. Report the results in the usual form. Does it make sense to hold both $expendA$ and $expendB$ fixed, while changing $shareA$? How would you interpret the coefficient on $shareA$? *Hint: Notice that $shareA = 100 \frac{expendA}{expendA + expendB}$ so when you change, say $expendA$, $shareA$ also moves, and similarly when you change $expendB$.*
- (f) In the model from part (e), find the partial effect of $expendB$ on $voteA$, holding $prtystrA$ and $expendA$ fixed. Evaluate this at $expendA = 300$ and $expendB = 0$ and compute a 95 confidence interval around this partial effect. Comment on the results. *Hint: Notice that $shareA = 100 \frac{expendA}{expendA + expendB}$, so when computing the partial effect³ of $expendB$ you will need to take this into account.*
- (g) Now consider the following specification of the model

$$voteA = \gamma_0 + \gamma_1 prtystrA + \gamma_2 \log(expendA) + \gamma_3 \log(expendB) + u.$$

What is the interpretation of γ_2 ?

- (h) In terms of the parameters, state the null hypothesis that a 1% increase in A's expenditures is offset by a 1% increase in B's expenditures.
 - (i) Estimate the model and report the results in the usual form. Do A's expenditures affect the outcome? What about B's expenditures?
 - (j) Test the hypothesis from part (2h) using a two-sided alternative at the 5% and 1% confidence levels. What do you conclude?
 - (k) Which model between (2a) and (2g) do you prefer? Please give *sensible arguments* when answering this question.
3. The data set NBASAL contains salary information and career statistics for 269 players in the National Basketball Association (NBA).
- (a) Estimate a model relating points-per-game ($points$) to years in the league ($exper$), age , and years played in college ($coll$). Include a quadratic in $exper$; the other variables should appear in level form. Report the results in the usual way.
 - (b) Holding college years and age fixed, at what value of experience does the next year of experience actually reduce points-per-game? Does this makes sense?
 - (c) Why do you think $coll$ has a negative and statistically significant coefficient? (*Hint: NBA players can be drafted before finishing their college careers and even directly out of high school.*⁴)
 - (d) Add a quadratic in age to the equation. Is it needed? What does this appear to imply about the effects of age, once experience and education are controlled for?

³Partial effect means partial derivative.

⁴Like was the case for Kobe Bryant.

- (e) Now regress $\log(\text{wage})$ on *points*, *exper*, exper^2 , *age* and *coll*. Report the results in the usual format.
- (f) Test whether *age* and *coll* are jointly significant in the regression from part (3e). What does this imply about whether age and education have separate effects on wage, once productivity and seniority are accounted for?
4. What is the effect of physical attractiveness on wage? Hamermesh and Biddle (1994)⁵ used measures of physical attractiveness in a wage equation. In this exercise you will use the file BEAUTY which contains a subset of the variables (but more usable observations) reported by Hamermesh and Biddle (1994). Each person in the sample was ranked by an interviewer for physical attractiveness, using five categories (homely, quite plain, average, good looking, and strikingly beautiful or handsome). Because there are so few people at the two extremes, the authors put people into one of three groups for the analysis: average, below average and above average, where the base group is *average*.
- (a) Using the data pooled for men and women, estimate the equation

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{belavg} + \beta_2 \text{abvavg} + \beta_3 \text{female} + \beta_4 \text{educ} + \beta_5 \text{exper} + \beta_6 \text{exper}^2$$

and report the results in the usual way. Are any of the coefficients surprising in either their signs or magnitudes? Is the coefficient on *female* practically large and statistically significant?

- (b) Explain in words what the hypothesis $H_0 : \beta_1 = 0$ against $H_a : \beta_1 < 0$ means and find the *p*-value.
- (c) Add interactions of *female* with all other explanatory variables in the equation from part (4a) (five interactions in all). Compute the *F*-test of joint significance of the five interactions. What do you conclude?
- (d) In the full model with interactions, determine whether those involving the looks variables—*female*×*belavg* and *female*×*abvavg*—are jointly significant. Are their coefficients practically small?
- (e) Is there convincing evidence that women with above average looks earn more than women with average looks? Explain.

⁵Hamermesh, Daniel S., and Jeff E. Biddle. "Beauty and the Labor Market." The American Economic Review 84, no. 5 (1994): 1174-194. <http://www.jstor.org/stable/2117767>.